

# ULTRASOUND WATERMARK: Real-time Acoustic Watermarking against Voice Scams on Smartphones

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**Abstract**—In this paper, we present ULTRASOUND WATERMARK, a real-time, smartphone-native framework for protecting live phone calls against voice impersonation attacks. Recent advances in text-to-speech (TTS), voice conversion (VC), and other AI-driven speech synthesis attacks enable adversaries to convincingly mimic trusted individuals during phone calls, creating a growing threat of voice scams. To defend against such attacks, we introduce a real-time acoustic watermarking system that binds speech to a caller’s physical articulatory motion. During a call, the caller’s smartphone continuously emits ultrasonic signals and captures reflections from mouth, jaw, and cheek movements using only the built-in speaker and microphone. The ultrasound signal is physically and temporally coupled to audible speech production, making it difficult to synthesize or spoof without the underlying articulatory movements. Our system is fully deployable on commodity smartphones and requires no additional hardware or explicit user interaction. We implement ULTRASOUND WATERMARK on both iOS and Android devices and collect a synchronized speech-ultrasound dataset containing over 2.5 hours of recordings from 22 speakers across eight smartphones and four real-world environments. Experimental results demonstrate strong robustness against replay, synthesis, and articulation-mimicking attacks under common telephony distortions while preserving perceptual audio quality, highlighting the practicality of real-time protection for voice communication.

## I. INTRODUCTION

Voice is the most natural form of human communication, carrying not only linguistic information but also biometric cues unique to each speaker. In mobile environments, the human voice increasingly serves as an identifier for copyright, content protection, data access, and secure communication. However, the rapid advancement of generative AI has significantly altered the threat landscape. Modern Text-to-Speech (TTS) and Voice Conversion (VC) systems, such as WaveNet-based and GAN-based models [9], [10], [17], [20], can produce synthetic voices that closely mimic real individuals with minimal training data. This development has fueled the rise of impersonation-based attacks such as voice phishing (“vishing”), which caused U.S. consumers more than \$2.95 billion in losses in 2024 alone [6], [16]. Studies show that humans are poorly equipped to detect synthesized voice clones, often performing at chance levels when asked to distinguish them from genuine speech [7], [23].

**Scenario.** Consider a scenario where a user receives a call that appears to come from someone they know, such as a family

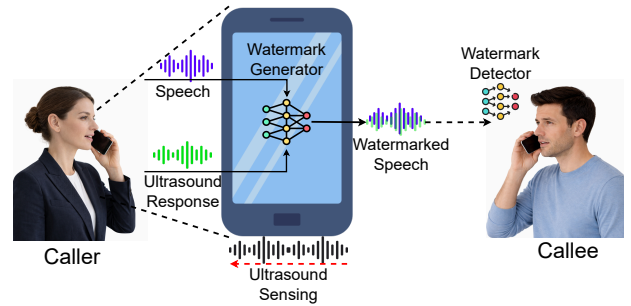


Fig. 1: ULTRASOUND WATERMARK working scenario.

member or colleague. The caller may pose as a supervisor and request urgent actions like approving a payment, or transferring funds. Because the voice sounds familiar, the callee may trust the call and act quickly. A successful attack can cause financial loss, data leakage, and long-term damage to trust and professional relationships. In this setting, the callee must decide in real time whether the received speech is coupled to contemporaneous live articulation rather than being replayed or synthesized.

**Existing Efforts.** As illustrated in Table I, prior defenses fall short in this scenario. *Deepfake Detectors*, such as ASVspoof-style systems [25]–[27], focus on analyzing audio content to detect its synthesized artifacts. While they can defend against replay and synthesis attacks, acoustic artifacts they rely on are often suppressed or distorted in real phone calls due to telephony codecs, echo, and streaming effects. As a result, they are poorly suited to defend against impersonation during live calls. *Digital Watermarking* embeds imperceptible bits into legitimate speech, allowing the watermark to be robustly extracted or detected for authentication. Since watermark embedding and verification are typically carried out offline, these methods generally cannot verify that the speech is coming from a legitimate speaker in real time during a call [3], [13], [18], [21], [31]. *Liveness Detection* verifies that the audio originates from a live human speaker, and typically requires specialized sensors or explicit user interaction [8], [11]. However, liveness alone does not establish identity. An attacker may pass liveness by speaking into the phone while injecting or replaying synthesized speech to impersonate a

| Approach                                | In-call | No Extra HW | Phone Native | Replay Attack | AI Synthesis (TTS/VC) | Live Human Impersonation | User Effort | Example References          |
|-----------------------------------------|---------|-------------|--------------|---------------|-----------------------|--------------------------|-------------|-----------------------------|
| Deepfake Detectors (ASVspoof, ML)       | △       | ✓           | △            | △             | ✓                     | ✗                        | △           | [25]–[27]                   |
| Audio Watermarking                      | ✗       | ✓           | △            | ✗             | △                     | ✗                        | ✓           | [3], [13], [18], [21], [31] |
| Liveness Detection (sensors, protocols) | △       | Often       | △            | ✓             | △                     | ✗                        | △           | [11], [19], [29]            |
| Provenance (platform-level)             | ✗       | ✓           | ✓            | ✗             | △                     | ✗                        | ✓           | [4], [15]                   |
| <b>Our work: ULTRASOUND WATERMARK</b>   | ✓       | ✓           | ✓            | ✓             | ✓                     | ✓                        | Low         | -                           |

TABLE I: Comparison of defenses against voice impersonation for live calls on commodity smartphones.

trusted person, for example, a family member or supervisor [11], [19], [29]. Finally, *broadier provenance initiatives and platform-level policies* support media origin tracking, but they do not cryptographically or physically bind the audio to a legitimate human speaker in an ongoing phone call [4], [15]. Collectively, these gaps point to the lack of a practical in-call defense that can defend against attacks under real-time conditions, establish liveness, and confirm speech veracity.

To defend against in-call impersonation attacks, we propose a real-time, smartphone-native legitimate caller verification system, named **ULTRASOUND WATERMARK**, which embeds watermarks tied to a caller’s physical articulation. It includes two mobile applications: a watermark generator on the caller’s device and a watermark detector on the callee’s device. As shown in Fig. 1, during a call, the caller’s phone continuously emits ultrasound and records reflections from orofacial motion (i.e., the mouth, jaw, and cheeks) using the built-in speaker–microphone pair. These articulation-informed signals are fused with the outgoing speech to produce a live, dynamic watermark. The watermark is imperceptibly embedded into the speech and then transmitted to the callee as a watermarked speech stream. On the callee side, a lightweight detector verifies the received signal along three layers:

- **Protocol Compliance:** The detector verifies that the received signal conforms to the expected watermarking pipeline, ensuring it is compatible for watermark decoding. Signals that do not satisfy these constraints are rejected before further verification.
- **Live Speech Presence:** The detector verifies the presence of real-time speech activity near the device by detecting ultrasound reflections induced by articulatory motion in conjunction with the watermark generator. This confirms active human speech production near the phone.
- **Speech-Motion Consistency:** Rather than enroll, identify, or authenticate a speaker, the detector verifies whether the received speech with generated watermark is causally consistent with the contemporaneous articulatory motion. Replay and synthesis attacks that lack sufficiently matched contemporaneous articulatory motion are expected to fail this check because their audible and ultrasonic components do not exhibit the required cross-modal consistency.

We highlight our contribution as follows:

- **Real-time Ultrasound Watermarking Framework:** The system emits ultrasonic signals and records articulatory reflections. It extracts phase and amplitude features from these reflections and fuses them with audible speech to

generate a frame-wise watermark tied to live speech production. A synchronized generator–detector architecture learns temporal alignment and is designed to distinguish genuine watermarked speech from replay, synthesis, and ultrasound spoofing combinations.

- **Multi-condition Evaluation:** We collect and release a synchronized speech–ultrasound dataset comprising over 2.5 hours of recordings from 22 speakers, captured using 8 commodity smartphones across four real-world environments. We consider three attack scenarios: audio-only spoofing, random ultrasound spoofing, and speech-synchronized ultrasound spoofing attacks. For each spoofing scenario, we evaluate two types of spoofed audio: replay and synthesized speech. We also evaluate robustness under common telephony distortions and codecs. The results demonstrate robustness across a range of attacks and distortions, with limited impact on perceptual audio quality.
- **Cross-Platform Mobile System:** We implement a complete prototype on both iOS and Android smartphones. The system runs fully on device and requires no extra hardware. Using the prototype, we demonstrate real-time watermark generation and detection during live calls. We also show how the system handles all three attack scenarios in practice. The application demo can be found at <https://ultrasoundwatermark.github.io>.

## II. THREAT MODEL

**User Setting.** The victim is a security-conscious human callee who *makes security-sensitive decisions based on perceived speech*, and our framework is deployed on both caller and callee devices. We assume the victim uses our app for calls involving financial or sensitive data, and expects the other party to use the same app. This setting is practical because secure calling is typically provided by an application-level service that requires both endpoints to adopt the same platform (e.g., WhatsApp [24] and Apple FaceTime [1]).

**Attacker’s Goal.** The attacker aims to impersonate a trusted speaker (e.g., a family member or supervisor) during a call and induce the callee to take a security-sensitive action under false pretenses. Concretely, the attacker attempts to cause the callee-side verification framework to accept attacker-controlled speech as containing a valid watermark, thereby making the call satisfy the consistency checks and increasing the likelihood that the callee complies with the request.

**Attacker Capabilities.** We assume the attacker can place real-time calls using smartphones and can obtain prior voice samples of the target speaker (e.g., from social media or

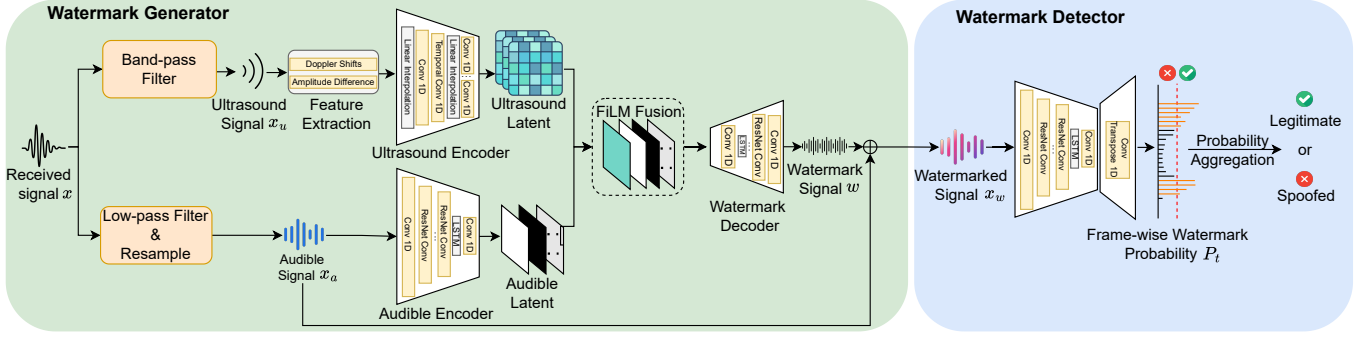


Fig. 2: Overview of ULTRASOUND WATERMARK.

previous calls) to mount replay or synthesized speech attacks. The attacker may also possess their own device running the ULTRASOUND WATERMARK app, but cannot remotely access the target call’s live ultrasonic sensing path or reproduce the target speaker’s ultrasound response coupled to the acoustic path. Thus, a synthetic or replayed audible signal may imitate the target speech, but it does not automatically reproduce the synchronized audio–ultrasound coupling observed during a live call. We examine three attack scenarios: targeting the audible channel alone (Scenario 1), attempting to manipulate both the audible speech and ultrasonic modality using random or speech-synchronized ultrasound (Scenarios 2 and 3).

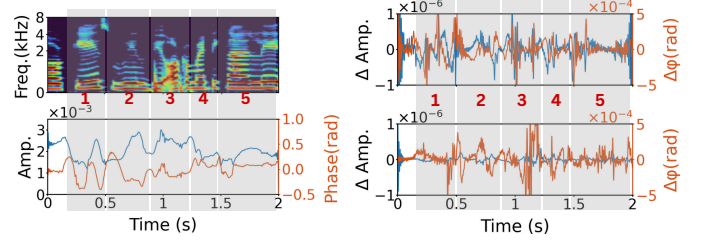
### III. SYSTEM DESIGN

We propose ULTRASOUND WATERMARK, a dual-modality framework that embeds articulatory cues from ultrasonic reflections into speech for real-time legitimate caller verification on smartphones. As shown in Fig. 2, the system consists of a *Watermark Generator* and a *Watermark Detector*. The generator jointly encodes audible speech and ultrasonic signals, fusing speech latents with the ultrasound feature map that captures instantaneous phase and amplitude variations. This design grounds the watermark in live articulatory motion rather than acoustic replay. The detector performs lightweight on-device inference, estimating frame-level watermark confidence and aggregating it over time for integrity verification. All components are optimized for low latency, robustness, and smartphone deployment, enabling continuous, imperceptible legitimate caller verification in real-world settings.

#### A. Ultrasound Feature Extraction and Fusion

We propose a lightweight design that deliberately balances expressiveness, robustness, and real-time deployability on commodity smartphones. We focus on extracting a minimal yet sufficient set of motion cues that are (i) causally tied to speech production, (ii) difficult to spoof via replay or synthesis, and (iii) efficient for continuous in-call operation.

**Design Rationale.** Our design is guided by three insights. First, articulatory motion occurs at millimeter and millisecond scales that low-rate frame-based features (e.g., 10–20 ms windows) fail to capture. Continuous-wave ultrasound enables



(a) Speech vs. ultrasound features. (b) Legitimate vs. spoofed.

Fig. 3: Ultrasound features for speech. Shaded regions correspond to the spoken words, 1: “Your”, 2: “case”, 3: “will”, 4: “be”, 5: “automatically”. (a) Speech Mel spectrogram with ultrasonic amplitude and phase features. (b) Phase and amplitude differences for legitimate ultrasound (Top) versus speech-synchronized spoofed ultrasound (Bottom).

fine-grained sample-level sensing naturally aligned with phonetic events [22], [30]. Second, spoof resistance requires physically grounded features. While absolute spectral patterns are easily replayed or synthesized, temporal derivatives of phase and amplitude induced by live motion are significantly harder to forge. Third, mobile constraints rule out heavyweight and non-causal processing, necessitating an incremental pipeline built on lightweight signal processing.

During a call, the caller’s smartphone continuously emits a multi-carrier ultrasonic signal [30]. We choose a base frequency of  $f_0 = 20$  kHz with six equally spaced carriers ( $\Delta f = 300$  Hz). The transmitted signal can be represented as  $u(t) = \sum_{i=0}^5 \cos(2\pi(f_0 + i \times \Delta f)t)$ , where  $f_0 + i \times \Delta f$  denotes the frequency for the  $i$ -th carrier. Meanwhile, the smartphone records the received signal  $x(t)$  at a sampling rate of 48 kHz using its built-in microphone, which contains both the ultrasound reflection  $x_u(t)$  and the audible speech  $x_a(t)$ . Both ultrasonic reflections and audible speech originate from the same articulatory motion, providing natural temporal alignment without requiring external synchronization.

**Ultrasound Feature Extraction.** We extract ultrasonic features by isolating each carrier with lightweight Finite Impulse Response (FIR) band-pass filters and demodulating the signal to baseband using cosine–sine (I/Q) mixing [2], [12], [28]. This enables sample-level tracking of instantaneous phase and amplitude that captures fine-grained articulatory motion natu-

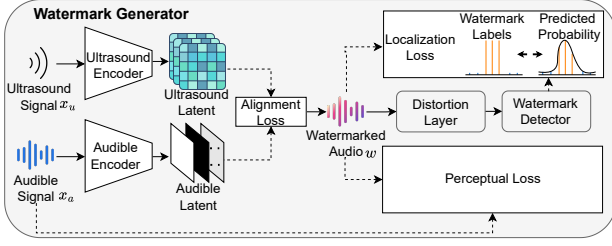


Fig. 4: Watermark generator-detector training pipeline.

rally aligned with speech, as illustrated in Fig. 3a. To improve robustness against spoofing, we compute temporal differences of phase and amplitude to emphasize non-stationary motion cues while suppressing static channel effects, with clear separation between genuine and spoofed signals shown in Fig. 3b.

**Audible Speech Feature Extraction and Encoding.** In parallel, the received signal is low-pass filtered and resampled from 48 kHz to 16 kHz using lightweight FIR filtering and linear interpolation. This aligns the audio representation with telephony codecs and minimizes perceptual distortion during watermark embedding. The audio pipeline is intentionally simpler than the ultrasound pipeline, reflecting our design choice that ultrasound provides the primary liveness signal, while audible speech serves as the perceptual carrier.

The resampled audio is then encoded using a strided residual convolutional network inspired by Encodec [5]. This encoder progressively reduces temporal resolution while increasing channel dimensionality, producing a compact latent representation that preserves the phonetic and prosodic structure necessary for reliable watermark insertion.

### B. Adversarially-aligned Training

We design an adversarially-aligned training framework that forces the generator-detector pair to learn the characteristic synchrony between speech and ultrasonic articulatory motion. Fig. 4 summarizes our approach. Paired audible speech and ultrasound signals are first routed through the watermark generator, which produces a watermarked audio. A synchronized distortion layer applies codec compression, echo, and reverberation to watermarked audios, ensuring that the model regularly encounters realistic phone call degradations and spoof-like perturbations. The resulting watermarked audio is then processed by the Watermark Detector, which outputs frame-level watermark probabilities. During training, speech is treated as a genuinely multimodal event in which acoustic sound waves and the sensed ultrasound are in lockstep. The model is trained so that only this co-evolving pattern triggers watermark activation, while any disruption of synchrony caused by replay attacks, mismatched motion, silent mimicry, or channel distortions is explicitly penalized.

**Joint Generator-Detector Optimization.** We jointly train the watermark generator and detector in an end-to-end fashion. The generator embeds watermarks into both genuine audios (with aligned audio-ultrasound pairs) and spoofed audios (with misaligned pairs), while the detector learns to distinguish be-

tween them. This joint optimization ensures that the watermark encoding captures temporal synchrony information that the detector can reliably verify at inference time.

**End-to-end Training and Detection.** During training, the generator receives the raw waveform  $x$  and produces both a watermark signal  $w$  and an extracted audible speech  $x_a$ . The watermarked audio  $x_w$  is constructed as:  $x_w = x_a + \alpha w$ , where  $\alpha$  is a scalar controlling watermark strength. To model realistic channels, distortions are applied to both  $x_w$  and  $x_a$ . Applying perturbations symmetrically to both clean and watermarked signals ensures that the detector learns invariances to real-world degradations, rather than relying on fragile spectral artifacts.

Gradients from the localization loss, alignment loss, and perceptual losses are backpropagated through the entire generator-detector pipeline. Gradient norms are monitored for stability, and gradient clipping together with a balancer mechanism prevents any single objective from dominating training. These create a training process that is stable, resource-efficient, and directly optimized for accuracy, robustness, and imperceptibility under real-world acoustic and adversarial conditions.

### C. Real-time Implementation

We design a real-time streaming architecture that enables the generator and detector applications to operate continuously during live phone calls under strict latency and computational constraints. The generator processes the audio stream in fixed increments, producing short watermarked segments in real time. Simultaneously, the detector aligns its sliding window with the generator to yield low-latency, frame-wise confidence estimates. This design transforms what would otherwise be a high-latency neural pipeline into a fully streaming system, achieving minimal latency overhead while maintaining stable watermark generation and detection throughout the call.

### D. Commodity Device Deployment

We deploy our system end-to-end on commodity mobile devices via native iOS and Android applications. The generator and detector models are exported to platform-compatible formats, enabling on-device inference without external servers. Communication between devices is established through IP-based calling, which streams the watermarked speech from the caller device to the callee device.

**Cross-Platform On-Device Inference.** To ensure broad compatibility, the generator and detector are converted to lightweight formats. The Android app embeds a model accelerated by NCNN [14], while the iOS app uses a Core ML variant optimized for real-time audio pipelines. Both applications run a fully local inference loop for generation or detection, avoiding cloud services and preserving privacy. In addition, many standard signal processing libraries such as `torchaudio` are unavailable or inefficient on phones. We implement mobile-compatible alternatives including lightweight FIR filtering, linear resampling, and exports, which allow the system to run natively on both Android and iOS without additional hardware.

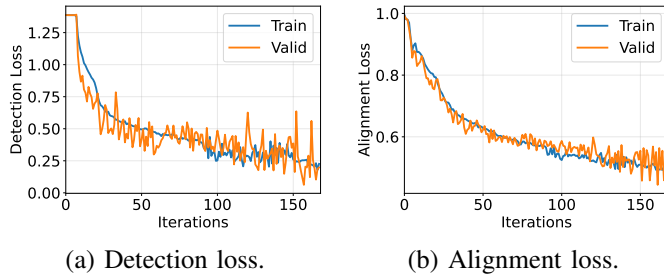


Fig. 5: Training loss over iterations.

**Instant Confidence Estimation.** When performing detection in a real-time phone call, the detector produces both an *instant confidence level* and a stabilized *averaged confidence level*. The instant confidence reflects the model’s output for the current moment, which is essential for frame-level responsiveness but may fluctuate due to momentary noise or transient distortions. Both confidence levels are displayed on the detector app, so that a sustained drop in the confidence readings visibly warns the user that the ongoing call is not authentic.

#### IV. EVALUATION

##### A. Experiment Setting

**Datasets.** To train and evaluate our framework, we construct a comprehensive dataset consisting of both genuine and attack samples, totaling over 2,000 sentence samples (over 2.5 hours of speech). The dataset is randomly split at the utterance level into training, validation, and evaluation sets using an 80%/10%/10% ratio. During training, all audio is segmented into two-second clips at a sampling rate of 48 kHz. The **genuine dataset** contains over 1,700 sentence samples from 22 participants, collected using purpose-built Android and iOS apps that simultaneously emit ultrasonic signals and record synchronized speech and ultrasound reflections on commodity smartphones. Recordings are captured on four iPhones and four Android smartphones across diverse environments (home, office, lab, and outdoor) under both speaker and earpiece call modes. Data from all devices and modes are combined during training. The **attack dataset** covers three representative threat scenarios: (1) audio-only spoofing, including replay, TTS, and VC attacks; (2) random ultrasound spoofing, where unrelated ultrasound signals are injected alongside synthesized speech; and (3) speech-synchronized ultrasound spoofing, where ultrasound signals are taken from a different speaker uttering the same sentence. In total, 300 synthesized samples are generated and dynamically paired with genuine recordings to simulate realistic attack conditions.

**Hardware.** The model is trained on an NVIDIA RTX 5090 GPU with 32 GB of memory. After training, the generator and detector are deployed on commodity smartphones for on-device inference, with the generator on the caller side and the detector on the receiver side.

**In-call Distortions.** To test robustness under realistic phone-call conditions, we apply various audio distortions to the watermarked audio before detection. All distortions are applied

at a sample rate of 16 kHz. These include filtering, resampling, compression, noise injection, gain changes, smoothing, echo, and neural audio coding.

**Evaluation Metrics.** We evaluate both watermark detection performance and speech quality. Detection performance is measured using Accuracy, True Positive Rate (TPR), False Positive Rate (FPR), Attack Success Rate (ASR), and the Receiver Operating Characteristic (ROC) curve with Area Under the Curve (AUC). Speech quality is evaluated using Perceptual Evaluation of Speech Quality (PESQ), Short-Time Objective Intelligibility (STOI), and Scale-Invariant Signal-to-Noise Ratio (SI-SNR).

##### B. Model Training Performance

We merge the Genuine Dataset and Attack Dataset to train the model. During training, we randomly sample from both datasets and feed them into the training pipeline in an end-to-end manner. For each training step, the watermark generator produces watermarked audio from genuine samples, while attack samples are used to produce negative examples. The watermark detector is trained to distinguish between these two cases. Our training objective minimizes a combination of losses. This encourages the generator to embed watermarks that are both detectable by the detector and robust to attacks, while remaining imperceptible in the audible speech.

**Loss Convergence.** Fig. 5 shows detection and alignment losses steadily decreasing and closely tracking validation, indicating stable optimization and limited train-validation gap.

**Audio Quality.** We evaluate whether embedding the watermark degrades perceptual audio quality. The watermarked audio achieves a PESQ score of 3.99, indicating near-transparent perceptual quality. An SI-SNR of 25.6 dB demonstrates that the watermark introduces relatively low signal distortion, and a STOI score of 0.922 confirms that speech intelligibility is effectively preserved.

**Watermark Detection Performance.** Fig. 6 shows detection performance over training. TPR and TNR increase steadily, with TPR reaching 0.93 and TNR 0.99. A 0.5 random baseline highlights that the model outperforms chance.

##### C. Performance Under Spoofing Attacks

We evaluate how our system defends against different spoofing attacks. Table II summarizes the detection performance across all scenarios, while Fig. 7 presents the corresponding ROC curves in terms of different watermark strength  $\alpha$ .

- **Audio-only Attacks:** For audio-only replay and synthesis attacks, the detector performs reliably, with AUC values of 1.00 and FPR of 0.01. Fig. 7 shows that ROC curves at all watermark strengths  $\alpha$  rapidly approach the upper-left corner. Increasing  $\alpha$  from 0.2 to 1.0 improves separability, with  $\alpha \geq 0.8$  achieving near-ideal detection.
- **Random Ultrasound Attacks:** When random ultrasound signals are combined with spoofed speech, detection performance remains perfect. Both replay and synthesis variants achieve an AUC of 1.00 with zero false positives, as reported in Table II. The ROC curves in Fig. 7 show

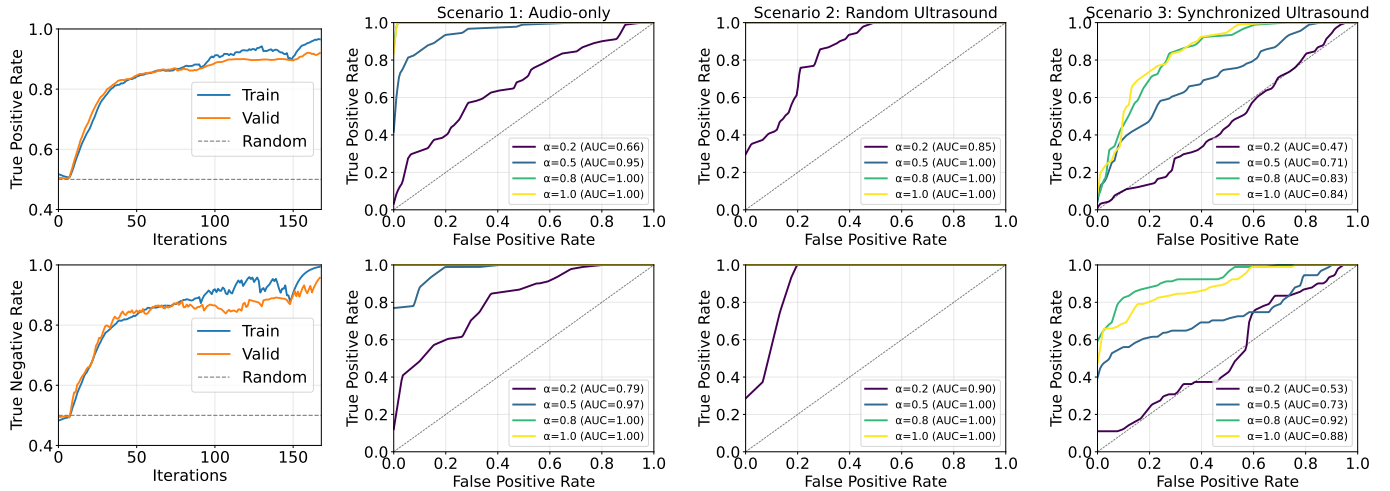


Fig. 6: Detection dynamics. Fig. 7: ROC of spoofing scenarios on replay (**Top Row**) and synthesis (**Bottom Row**) attacks.

| Attack Type                         | AUC         | FPR         |
|-------------------------------------|-------------|-------------|
| Audio-only + replay                 | 1.00        | 0.01        |
| Audio-only + synthesis              | 1.00        | 0.01        |
| Random ultrasound + replay          | 1.00        | 0.00        |
| Random ultrasound + synthesis       | 1.00        | 0.00        |
| Synchronized ultrasound + replay    | <b>0.84</b> | <b>0.37</b> |
| Synchronized ultrasound + synthesis | <b>0.88</b> | <b>0.30</b> |
| <b>Average</b>                      | <b>0.95</b> | <b>0.12</b> |

TABLE II: Detection results against attack types ( $\alpha = 1.0$ ).

that even at low watermark strength ( $\alpha = 0.2$ ), detection remains robust, indicating that unrelated ultrasonic patterns do not produce valid watermark signatures.

- **Synchronized Ultrasound Attacks:** Synchronized ultrasound attacks yield the lowest detection performance. As shown in Table II, the AUC drops to 0.84 for replay and 0.88 for synthesis, with FPR values of 0.37 and 0.30, respectively. Fig. 7 further shows a clear dependence on watermark strength. At a low  $\alpha$  (0.2), ROC curves are close to the diagonal, while increasing  $\alpha$  to 0.8 and 1.0 substantially improves detection performance. Despite the degradation, performance remains well above chance, indicating residual inconsistencies between audible speech and ultrasonic signals provide reliable cues for detection.

In summary, across all spoofing attack settings, our approach achieves strong performance against audio-only and random ultrasound attacks, even at low watermark strengths. Synchronized ultrasound attacks are more challenging, but detection improves consistently as the watermark strength  $\alpha$  increases.

We then evaluate the robustness of our watermark under two practical conditions: (i) **in-call distortions** and (ii) **VoIP codec transcoding**. These factors are inherent in real-world voice communication and can alter signal fidelity, potentially affecting watermark detectability. Table III reports representative results across both settings.

**In-call distortions.** As shown in Table III, the system remains

| Scenario                     | Setting                | TPR  | FPR  |
|------------------------------|------------------------|------|------|
| <b>Clean</b>                 | None                   | 0.93 | 0.01 |
| <b>In-call (Robust)</b>      | Noise / Resampling     | 0.93 | 0.01 |
| <b>In-call (Moderate)</b>    | Echo                   | 0.80 | 0.02 |
| <b>In-call (Challenging)</b> | High-pass Filter       | 0.02 | 0.04 |
| <b>Codec (High fidelity)</b> | G.711 / Opus (96 kbps) | 0.97 | 0.20 |
| <b>Codec (Low bitrate)</b>   | Opus (48 kbps)         | 0.08 | 0.50 |

TABLE III: Robustness under distortions and VoIP codecs.

robust under distortions such as noise and resampling, with a slight drop in performance (TPR  $\approx$  0.93, FPR  $\approx$  0.01). Moderate degradation appears under echo (TPR = 0.80), while aggressive operations such as high-pass filtering significantly reduce detection (TPR = 0.02) by removing low-energy watermark components. Overall, the system handles realistic distortions well but is sensitive to extreme signal manipulation. **VoIP codec transcoding.** High-fidelity codecs such as G.711 and high-bitrate Opus preserve performance (TPR  $\approx$  0.97), while aggressive compression degrades it. For example, low-bitrate Opus reduces TPR to 0.08 and increases FPR to 0.50. This reflects a trade-off between imperceptibility and robustness under strong compression.

## V. CONCLUSION AND FUTURE WORK

We present ULTRASOUND WATERMARK, a real-time, phone-native audio watermarking system that binds speech to articulatory motion using ultrasonic sensing. Evaluation shows robust protection against voice scams and in-call distortions while preserving audio quality. Future work will study imperceptibility, adaptive adversaries, and cross-platform performance on commodity smartphones.

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